THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 9: **Propositions and their Converses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The converse of "If a number is divisible by 6, then it is divisible by 3" is false because: 1

(a) 9 is divisible by 3 but not by 6

(b) two angles of 60° in an equilateral triangle are equal but not vertically opposite

(c) a kite that is not a rhombus has perpendicular diagonals

(d) $x = -3$ gives $x^2 = 9$ but $x \neq 3$

2. Which of the following is a counterexample to the statement "Every multiple of 3 is odd"? 1

(a) 9 **(b)** 15 **(c)** 3 **(d)** 6

3. For the expression $n^2 + n + 101$, the smallest natural number n for which the value is NOT prime is: 1

(a) 5 **(b)** 4 **(c)** 3 **(d)** 6

4. The number of factors of 72 is: 1

(a) 14 **(b)** 11 **(c)** 13 **(d)** 12

5. A triangle has sides 33 cm, 56 cm and 65 cm. Using the converse of the Baudhāyana–Pythagoras theorem, the triangle is: 1

(a) isosceles **(b)** right-angled **(c)** not right-angled **(d)** equilateral

6. Which statement is true for every integer n? 1

(a) $n(n+1)$ is even **(b)** $n+1$ is even **(c)** n^2 is even **(d)** $n(n+1)$ is odd

7. The sum of the 5 consecutive integers starting from 17 is: 1

(a) 90 **(b)** 96 **(c)** 95 **(d)** 100

8. The number of integers from 1 to 50 that have an odd number of factors is: 1

(a) 8 **(b)** 7 **(c)** 6 **(d)** 9

9. The converse of "If a quadrilateral is a square, then it is a rectangle" is false because: 1

(a) a kite that is not a rhombus has perpendicular diagonals

(b) $x = -3$ gives $x^2 = 9$ but $x \neq 3$

(c) two angles of 60° in an equilateral triangle are equal but not vertically opposite

(d) a 4 cm by 6 cm rectangle is not a square

10. Which of the following is a counterexample to the statement "If a number is divisible by 4 and by 6, then it is divisible by 24"? 1

(a) 72 **(b)** 24 **(c)** 48 **(d)** 12

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the converse of “If a number is a perfect square, then it has an odd number of factors” and test it. 2

12. Find the smallest natural number n for which $n^2 + n + 5$ is not prime, and the value obtained. 2

13. Is the number 121 a perfect square? Decide by counting its factors. 2

14. Write the converse of “If a triangle has two equal angles, then the sides opposite those angles are equal” and test it. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the converse of “If a triangle is right-angled, then the square of its longest side equals the sum of the squares of the other two sides”. Is the converse true? Justify. 3

16. Show that the statement “A number divisible by 3 and by 9 must be divisible by 27” is false by giving a counterexample. 3

17. If n is a multiple of 18, is n a multiple of 6? Write the converse and give a counterexample if false. 3

OR

Write the converse of “If a parallelogram is a rectangle, then its diagonals are equal”. Is the converse true? Justify. 3

18. Show that the statement “Every multiple of 3 is odd” is false by giving a counterexample. 3

OR

If n is a multiple of 24, is n a multiple of 8? Write the converse and give a counterexample if false. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that the sum of 3 consecutive even numbers starting at $2n$ is a multiple of 6. Check with $n = 4$. 5

OR

Take the two-digit number 47. Subtract the sum of its digits and show that the result is divisible by 9. Prove the same for any two-digit number. 5

20. Prove that the sum of 5 consecutive even numbers starting at $2n$ is a multiple of 10. Check with $n = 7$. 5

OR

Take the two-digit number 63. Subtract the sum of its digits and show that the result is divisible by 9. Prove the same for any two-digit number. 5

END OF QUESTION PAPER — Total: 40 Marks

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